

Assignment 7

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1) Using the variational principle and plane wave basis, determine the eigenvalues for the following square well potential: (10 marks)

$$V(x) = 0 \text{ for } x > b/2 \text{ and } x < -b/2$$

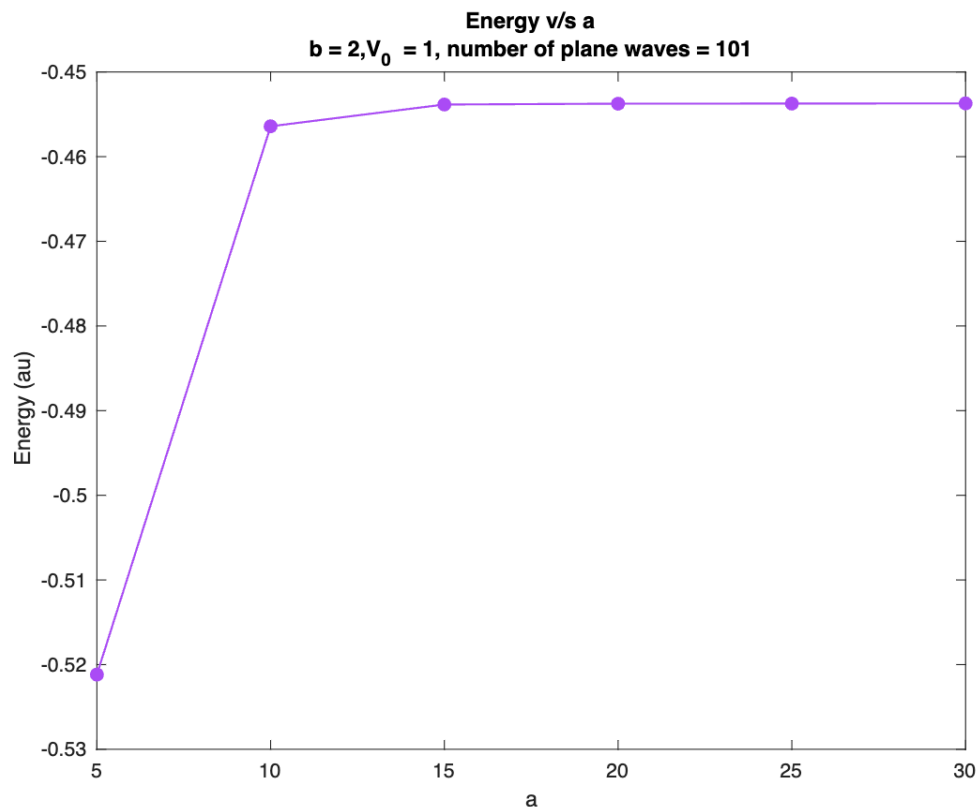
$$V(x) = -V \text{ for } -b/2 \leq x \leq b/2$$

Assume the plane waves have a periodicity a ; $a > b$. Use atomic units. Requires lapack dsyev. Answer the following questions:

(a) For $V=1$, $b=2$ what is the energy?

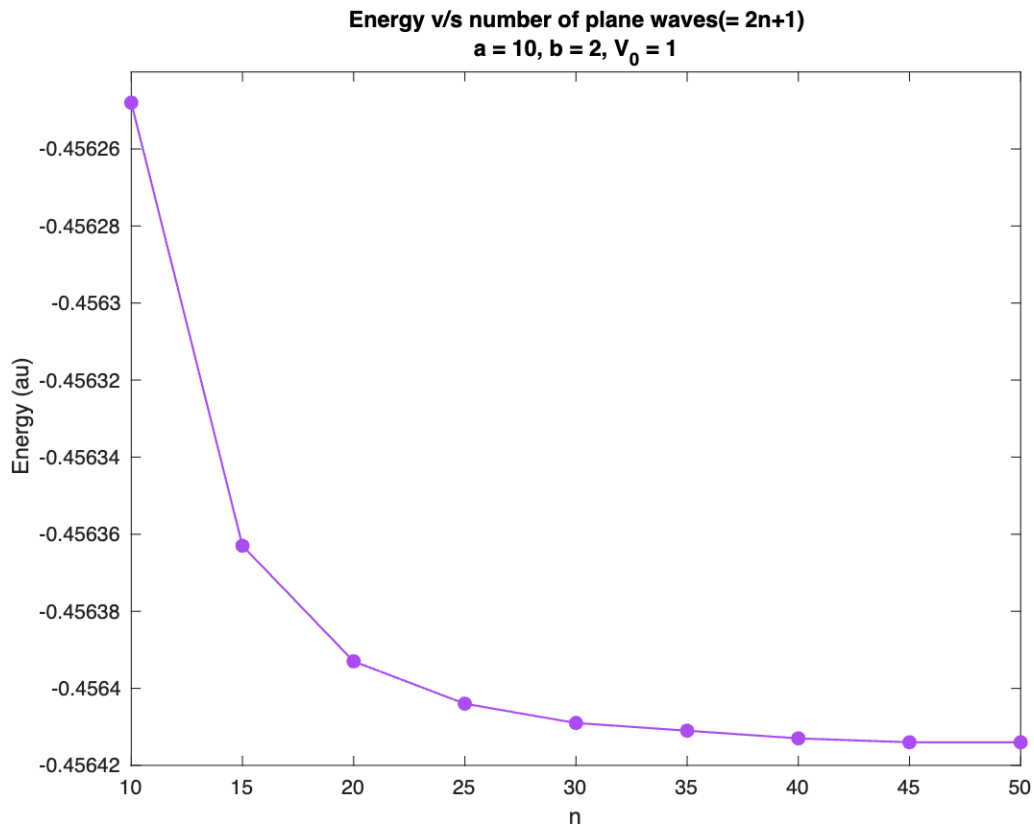
$$\text{Energy} = -0.453709 \text{ (} a = 30, n = 50 \text{)}$$

(b) Plot the energy as a function of ' a '. Explain what you observe.



We observe that the energy first increases with a and then stabilises asymptotically at around $a = 25$. This is because we are applying periodic boundary conditions and therefore what we are actually solving for is a series of potential wells. As we increase the value of a , the distance between the wells increases and since the potential is attractive, as the wells move further away from each other, the energy increases.

(c) Plot the energy as a function of no. of plane waves. What do you observe? Explain your observation.



We observe that the energy first increases with a and then stabilises asymptotically at around $n = 50$. This is because since we are applying the variational principle, and in variational principle our estimate of ground state energy is always greater than the actual value and as we increase the number of plane waves, we get a better approximation of the ground state energy.